ML Lab Practical 1

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Section: C - C2

Roll no: 29

In [1]:
```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

from sklearn.model_selection import train_test_split
```

In [2]:
```
df = pd.read_csv("titanic.csv")
```

In [3]:
```
df.head() #Dataframe.head
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Emb
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Emb
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	

In [4]:

```
df.tail()
```

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embark
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	

In [5]:

```
df.shape
```

Out[5]:

```
(891, 12)
```

In [6]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
```

```
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [7]:

```
df.describe()
```

Out[7]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [9]:

```
df.isnull().sum()
```

Out[9]:

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [10]:

```
df["Age"].fillna(df["Age"].median(), inplace = True)
```

In [11]:

```
df["Embarked"].fillna(df["Embarked"].mode()[0], inplace = True)
```

In [12]:

```
df.isnull().sum()
```

Out[12]:

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
```

```
Fare            0
Cabin         687
Embarked        0
dtype: int64
```

In [13]:

```
df.drop("Cabin", axis=1, inplace=True)
```

In [14]:

```
df.duplicated().sum()
```

Out[14]:

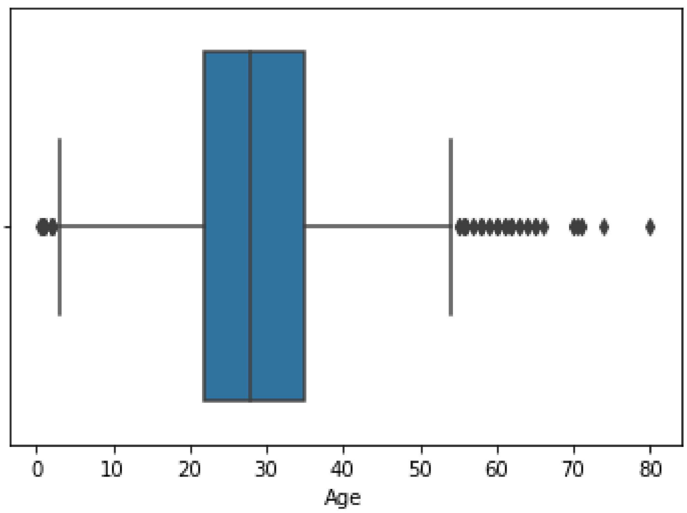
```
0
```

In [15]:

```
df.drop_duplicates(inplace=True) #optional: apply only when cell has duplicate values
```

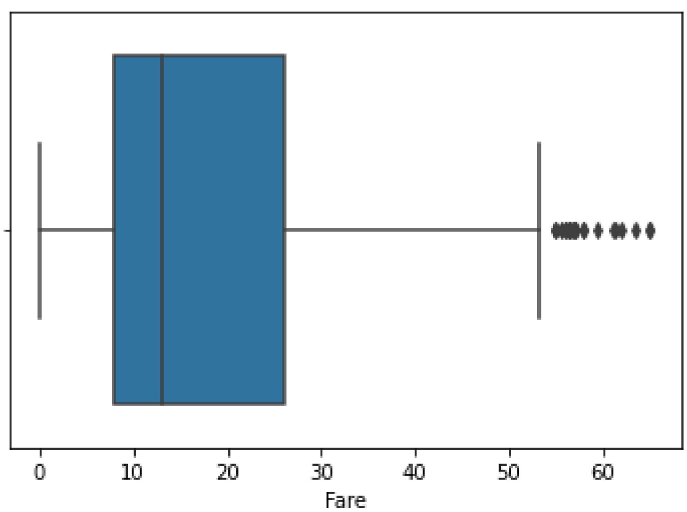
In [16]:

```
sns.boxplot(x=df["Age"])
plt.show()
```



In [23]:

```
sns.boxplot(x=df["Fare"])
plt.show()
```



In [22]:

```
Q1 = df["Fare"].quantile(0.25)
Q3 = df["Fare"].quantile(0.75)

IQR = Q3-Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df = df[(df["Fare"] >= lower) & (df["Fare"] <= upper)]
```

In [40]:

```
le = LabelEncoder()

df["Sex"] = le.fit_transform(df["Sex"])
```

```
C:\Users\HP\AppData\Local\Temp/ipykernel_9920/2594793548.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df["Sex"] = le.fit_transform(df["Sex"])
```

In [26]:

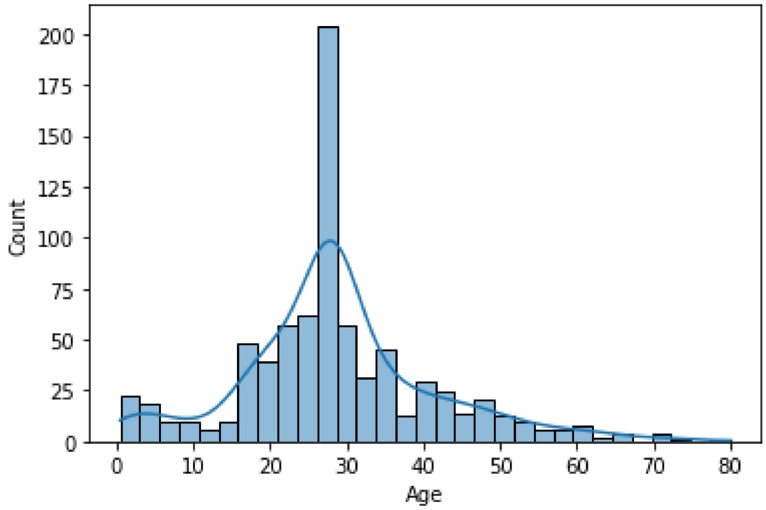
```
df.head()
```

Out[26]:

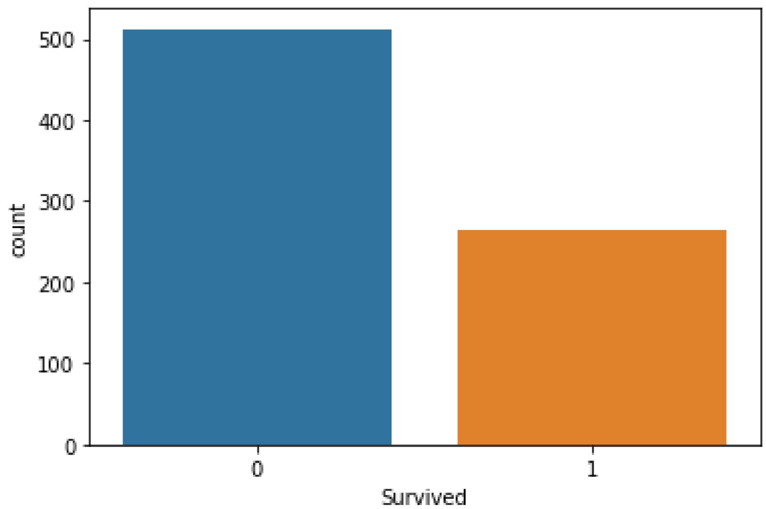
	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500	2
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250	2
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000	2
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500	2
5	6	0	3	Moran, Mr. James	1	28.0	0	0	330877	8.4583	1

In [28]:

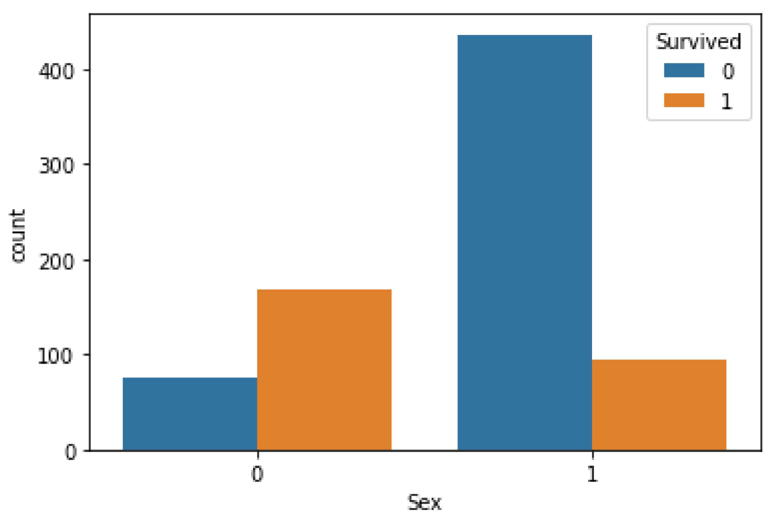
```
sns.histplot(df["Age"], kde=True)
plt.show()
```



In [29]:
```
sns.countplot(x="Survived", data=df) #univariant analysis : displaying using one value
plt.show()
```

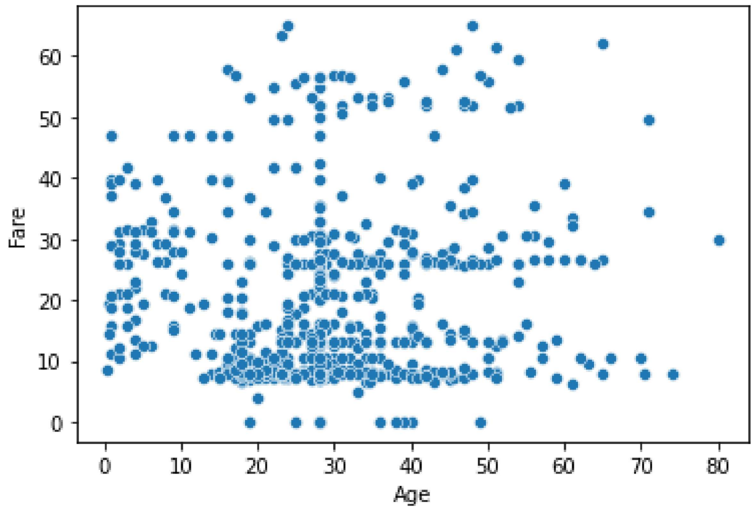


In [30]:
```
sns.countplot(x="Sex", hue="Survived", data = df) #bivariant analysis
plt.show()
```



In [31]:

```
sns.scatterplot(x="Age", y="Fare", data=df) #also used to detect outliers
plt.show()
```



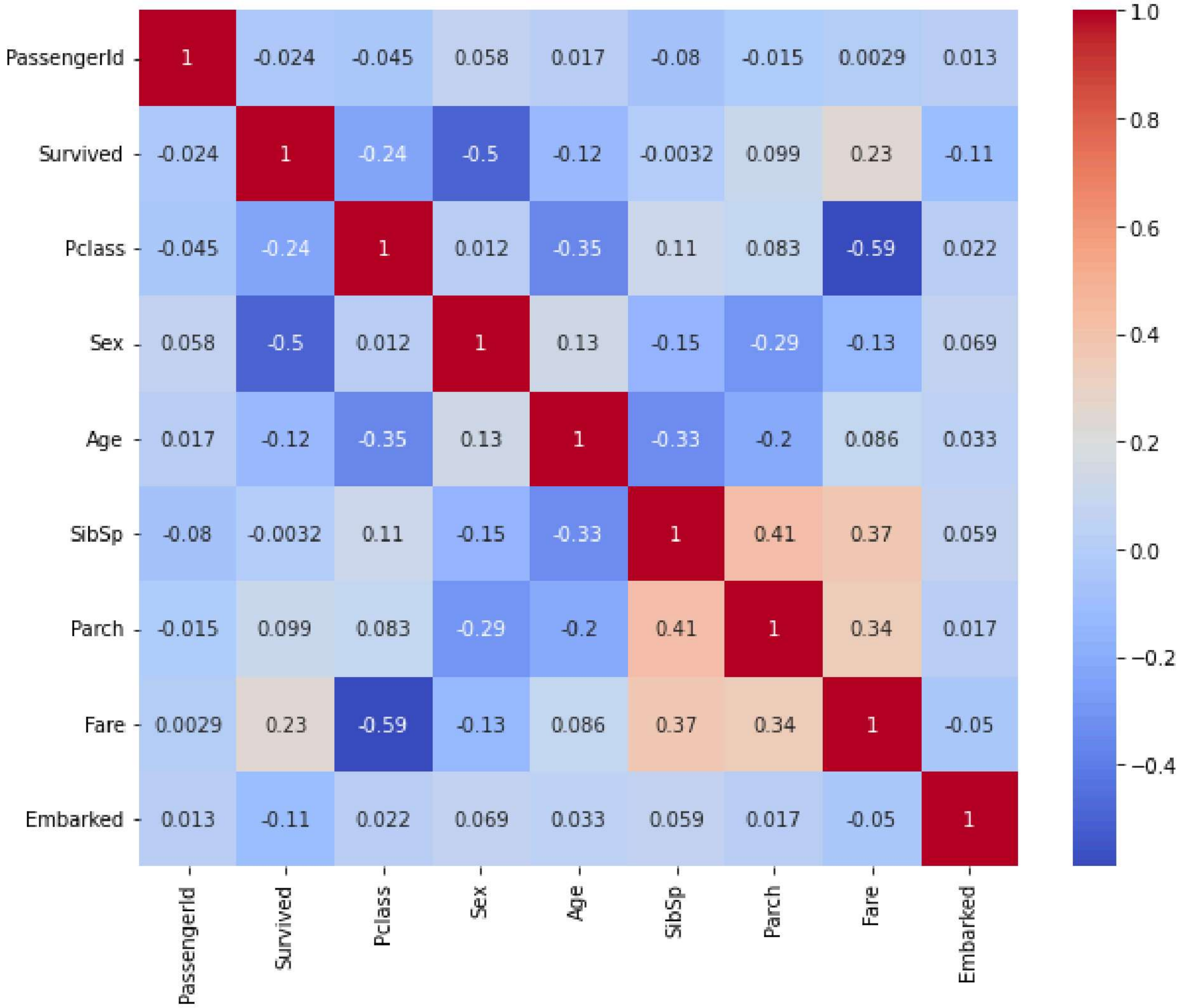
In [32]:

```
#multivariant analysis
plt.figure(figsize=(10,8))

numeric_df = df.select_dtypes(include=["number"])

sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')

plt.show()
```



In [35]:

```
scaler = StandardScaler()
df[["Age","Fare"]] = scaler.fit_transform(df[["Age","Fare"]])
```

```
C:\ProgramData\Anaconda3\lib\site-packages\pandas\core\frame.py:3678: SettingWithCopyWar
ning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_
guide/indexing.html#returning-a-view-versus-a-copy
  self[col] = igetitem(value, i)
```

In [36]:

```
X = df.drop("Survived", axis = 1)

y = df["Survived"]
```

In [37]:

```
X = X.drop(["PassengerId", "Name", "Ticket"], axis = 1)
```

In [38]:

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=
```

In [39]:

```
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(620, 7)
(155, 7)
(620,)
(155,)
```

In [ ]: